

Entropy-Enhanced Option Pricing: Improving Black-Scholes with Information Theory

Information Theory and Statistics - Project Report - Group 12

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1 Introduction to the concept of entropy

While the concept of entropy emerged historically from thermodynamics as a way of quantifying molecular disorder, its mathematical framework has found wide application in information theory, where it is known as Shannon entropy (Lairez, 2022). Shannon entropy measures the unpredictability of information. To better understand this formulation, it is important to examine the structure and rationale underlying its definition, particularly the logarithmic component and its weighting by probability. In information theory, the amount of information associated with a particular outcome is quantified using the notion of *self-information*:

$$I(x) = -\log_2 p(x)$$

The logarithmic form of self-information is justified by three fundamental properties:

1. **Non-negativity:** The information content is always non-negative, since for all $p \in (0, 1]$, we have $\log(p) \leq 0$, leading to $-\log(p) \geq 0$.
2. **Monotonicity and negative slope:** Rarer events convey more information, consistent with the derivative $(-\log(p))' = -\frac{1}{p} < 0$.
3. **Additivity:** For independent events, the total information is additive: $\log \prod p = \sum \log p$.

The use of base-2 logarithms corresponds to the binary encoding of information, where the entropy quantifies the expected number of binary (yes/no) decisions required to identify an outcome. For instance, an event with probability $\frac{1}{16}$ conveys $\log_2(16) = 4$ bits of information; this is equivalent to four binary queries needed to resolve the uncertainty.

Crucially, entropy extends beyond the information contained in a single event. It represents the *expected* information across all possible outcomes, where each outcome's information $\log_2 p(x)$ is weighted by its probability $p(x)$. This aligns with the law of large numbers and yields a probabilistically weighted average of information:

$$H(X) = - \sum_x p(x) \log_2 p(x)$$

Thus, entropy serves as a measure of the average uncertainty inherent in a random variable and captures the expected number of bits required to achieve optimal encoding or transmission. The more uniform the distribution of outcomes, the greater the entropy and hence the higher the underlying uncertainty.

Furthermore, entropy $H(X)$ is fundamentally linked to the amount of information a random variable conveys; it reflects the minimal number of bits needed to reconstruct the original message from its encoded binary form (Gray, 2023). In contrast, mutual information $I(X; Y)$ quantifies the reduction in uncertainty about one variable due to the knowledge of another, and is defined as the difference between marginal and conditional entropies. Notably, entropy can be viewed as a special case of mutual information, since $I(X; X) = H(X)$.

This study seeks to extend these information-theoretic concepts into the domain of financial theory. While the application of entropy within finance remains relatively underexplored compared to its use in physics or communication systems, there is a growing interest in evaluating whether entropy can serve as a complementary or even alternative metric to traditional measures such as volatility. Therefore, this prompts our first research question:

In which fields of financial theory has the concept of entropy been applied?

To be able to answer this research question, in particular, our paper will investigate the integration of entropy into mathematical finance, in particular option pricing models. Rather than adopting a full-fledged entropy pricing framework, our aim is to explore whether the inclusion of entropy can improve option pricing within conventional models.

In addition, we will aim to answer our second research question:

Can common option pricing techniques be improved with the inclusion of entropy?

To address these questions, our paper will introduce volatility as the heavily applied standard metric of financial uncertainty and compare it with entropy-based approaches. Following a brief overview of the existing literature and theoretical applications of information theory in finance, we will proceed to estimate entropy measures applicable to financial instruments. This estimate will be contextualized within the Black-Scholes framework, which is formally introduced as a baseline for comparison. Ultimately, we will contrast standard option prices with those derived from entropy-enhanced models to evaluate potential improvements in pricing accuracy and theoretical robustness.

2 Entropy and volatility in the stock market

As is well-known by statisticians and economists, there is no mathematical equation or algorithm that allows one to predict exactly how a stock price is going to evolve in the future. Because of this limitation, it is essential to study tools that can allow one to derive insights from the stock market so that the market's behavior becomes more predictable. All financial assets can be regarded as discrete random variables, as we are interested in their value expressed in currency units, which are not infinitely granular. Therefore, it suffices to look at entropy for discrete random variables.

Entropy in the stock market Starting with the definition of entropy as presented in the course Information Theory and Statistics, we have

$$H(X) := - \sum_x \log(p(x))p(x) = \sum_x \log(p^{-1}(x))p(x)$$

As mentioned previously, we can interpret entropy as the “expected value of the surprise” (Kharagpur, 2020) where unlikely events contribute a large value via $p^{-1}(x)$. While an individual unlikely event can significantly increase the entropy, the term is weighted by probability, and entropy as the uncertainty in the market in the sense of a lack of predictability is maximal when every event has the same likelihood (in other words, in the uniform case) (Sheraz et al., 2015). Entropy also has “ability to capture disorder or uncertainty in the system without putting any constraint on the probability distribution function” (Taneja et al., 2019). Despite these advantages, there is not much application of entropy in financial theory.

Volatility in the stock market Standard deviation, the square root of the second centralized moment, and volatility have become synonymous in finance as its common mea-

sure. Interpreting the definition,

$$\sigma(X) := \sqrt{\frac{1}{|X|} \sum_{x \in X} (x - \mu_X)^2}$$

it represents the uncertainty in terms of deviation from the expectation, such that high values refer to large price fluctuations. Moreover, it is a convenient choice of uncertainty quantification as in long-established models like log-Normal market dynamics, the underlying distribution takes the variance as input. However, it is not a perfect measure as it is sensitive to outliers Bentes and Menezes, 2012, and practitioners introduce bias when estimating it by using the moment around the origin instead of the moment around the mean (Hull, 2014).

Comparison of entropy and volatility The standard deviation alone, as it is usually called in statistics, is an insufficient measure of uncertainty in the market (Bentes & Menezes, 2012). The same holds for entropy alone - it is itself not sufficient, and it tells a different kind of uncertainty. Volatility quantifies the magnitude of price or return fluctuations, while entropy measures the unpredictability of the distribution of outcomes. Volatility is usually used as is, whereas advanced measures based on entropy exist. For example, mutual information can provide insights into overall uncertainty from dependencies of individual randomness that volatility cannot yield (Bentes & Menezes, 2012). Unlike volatility, entropy “can [capture] the uncertainty and disorder in a time series without imposing any constraints on the theoretical probability distribution” (Bentes & Menezes, 2012). Additionally, since entropy refers to more than one single statistic, it is a more powerful tool.

Entropy and volatility have an inverse relation (A. Gupta, 2023) when it comes to their value. On the one side, while the entropy is maximal in the uniform case, the volatility of the probabilities would take on its minimal value of 0 in this case. On the other side, in times of stock market crashes, volatility is high as the magnitude of asset prices widens downwards, whereas the high probability of prices moving downwards lead to low entropy (Olbryś & Majewska, 2022). As entropy relates to probabilities

(the distribution of outcomes), not to outcomes directly, it is insensitive to scaling and shifting of the outcomes, whereas volatility changes quadratic when scaled - but it is also invariant to scaling. Also, entropy relates to the probability mass or density function, while variance can be calculated for these, the realisations of a random variable and in general any numerical data set. While both measures relate to uncertainty, they are quite different in nature.

3 Entropy applications in finance

It is natural to apply measures of uncertainty to financial markets. They are part of all developed capitalist market-based economies, and they are after all, complex adaptive systems driven by diverse actors with heterogeneous expectations and information sets. These markets' inherent unpredictability and volatility make them well-suited for analysis using the concept of entropy, which quantifies disorder and uncertainty. Financial markets are critical to the functioning of modern economies as they allow for allocation of capital, sharing of financial risk and price discovery. Understanding and managing risk within these markets is paramount, not only for individual investors but also for the stability of the broader financial system. Entropy, with its ability to capture the inherent randomness and information content of market dynamics, offers a powerful framework for modeling and interpreting financial phenomena, providing valuable insights into pricing, risk management, portfolio construction, and trading patterns. This chapter will give an overview of the development and current status of research on these various applications of information theory in finance.

3.1 Pricing

Entropy pricing theory uses entropy to derive (risk-neutral) densities for stocks (Gulko, 1999) and bonds (Gulko, 2002). Under a risk-neutral density, as no-arbitrage pricing theory tells, the expectation of a payoff function yields a price for a derivative. For stock options, ETP results in a formula similar to Black-Scholes equation, but with less restrictive assumptions to derive it. The prices derived for option via ETP can be very close to real market prices even in a simple model (Mooij, 2019). The EPT option pricing in works such as Yu, 2020, Rompolis, 2009, and Ardakani, 2022 can even yield highly accurate prices in a sense of close to prices observed in the market.

Another idea is to stick closer to the standard Black-Scholes formula approach by using the existing Black-Scholes model with volatility as an input, but to derive a better (closer to the not necessarily known correct) value for implied volatility with the help of entropy (Taneja et al., 2019).

3.2 Risk Management

From what is risk to how we measure it to why we manage it As a first idea of what risk is, people may think of uncertainty (or volatility, as it is usually called in finance), which is actually close to risk. But volatility is indifferent between negative and positive outcomes, and positive outcomes are nothing rational people should be worried about and hence not what we want to define as risk. We understand a risk as the possibility of an unwanted negative outcome, whose occurrence is unknown and whose realisation we are exposed to (Holton, 2004). In the following, we are interested in outcomes measurable in numeric terms only. Then, it makes sense to ask how associated risks can be measured properly.

Given that it is clear what the risk is, the questions of how to measure it arises. We demand a good risk measure to adhere to a couple of properties. First, it should be monotonous. An asset that will always produce worse outcomes must be declared as riskier. Second, it should be translation-invariant: non-random parts as part of a portfolio should reduce the risk by their value. Third, it should be linear or convex, but not concave to not punish diversification. Fourthly, it should be - at least positively - homogeneous: scaling the asset should also scale the risk to the same extent (Bäuerle & Rieder, 2017). The most well-known risk measure, value-at-risk, is a quantile of the loss distribution and fulfils almost all of these criteria. Risk measures that follow all axioms like expected shortfall build up on VaR, and there are further risk measures such as the Dutch risk measure and expectiles (Cai & Mao, 2020). Finally, risk management is about the identification, measurement, and mitigation of risk (Thomson Reuters Corp., 2024) given a tolerated level of risk, since returns are associated with risks, and investors demand returns.

Risk and entropy A major factor in risk management is correlation, which when positive leads to fatter tails of the loss distribution, which leads to greater risk than compared to an uncorrelated case. As the density is more spread out, risk measures increase as well. Beyond Pearson-type correlation, mutual information, or the directional version transfer entropy can be used as measures of relationship, without being restricted to a linear type of co-movement (Yon Shin Teo, 2019 and Liu et al., 2020). Entropic ways to measure risk involve the combination of standard statistical moments and information theory, for instance the mutual information between assets' return and volatility. Dedu and Toma, 2015 suggest in addition the direct combination of established risk measures and entropy.

Besides in the estimation of the loss distribution for many types of risk, it is particularly useful for credit risk. Credit default risk is based on three estimates, one of which is the probability of default (Hull, 2014) whose estimation Mhaskar, 2013 suggests to challenge with entropy, and Carta et al., 2020 demonstrate that entropy can also be used for model credit scoring. Such models show similar performance to machine-learning based models. Lastly, liquidity risk of a portfolio can be accounted for with the estimation of the speed of capital increment (Zhou et al., 2013), for which information theory can be used.

3.3 Efficient Market Hypothesis

In finance - as common in business studies and economics -, the analysis begins with assuming ideal conditions under which there are clear conclusions from the presumed main driving forces in a market. In financial markets, a common model is that all information is reflected in prices, which is called strong market efficiency (Fama, 1970). As a consequence, demand and supply for securities always lead to welfare-maximising equilibrium market prices, and there is no information advantage exploitable for excess profits for any investor. In a less strict form, known as semi-strong, only public information is assumed to be known by investors. In its weakest form, only historical price information is correctly processed by the people acting on financial markets. If this is

not the case, there is no form of market efficiency.

Research on the EMH In the last century, Fama, 1970 suggested that the semi-strong form best describes financial markets. In the first decade of the twenty-first century, there is at least support for a delay in information processing, leading to speed-caused or underreaction-caused inefficiency (Bouattour & Martinez, 2019). Another root of inefficiency is the irrational behaviour of market participants, which according to Malkiel, 2003 leads to short-time deviation from prices optimal based on available information. Market efficiency also varies with time and location, with inefficiencies being particularly present in developing countries' financial markets (Patra & Hiremath, 2022). There is, however, also a kind of economic support of the EMH when inefficiencies are not exploitable by investors in the sense of no profit remaining after accounting for the cost of trading (Tıtan, 2015). Thus, the EMH is generally accepted up to the point where costs are considered, and as Naseer and Tariq, 2015 notes, it is now regarded as valid in relative terms, with only limited support for its weak and semi-strong forms. Furthermore, violations of efficiency such as market anomalies are not reliable (Woo et al., 2020). Market efficiency would be challenged by being able to reliably forecast stock prices just with historic price information, which has not been the case yet reliably (Boya, 2013) and some researchers believe that market efficiency will persist in the foreseeable future (Pasricha & Dhanda, 2022).

The contribution of information theory on the EMH Information theory, however, has a clear answer on market efficiency. The paradox of Grossman and Stiglitz, 1980, states that perfectly efficient markets are impossible. Fittingly, Brouty and Garcin, 2022 show with an entropy-based test that markets are often inefficient as in principle, “entropy is maximised when the market becomes more efficient” (Chortane & Naoui, 2024). The paradox, however, does not contradict the previously mentioned results as market can get nevertheless close to efficiency. Generalised entropy incorporating the cost of information such as proposed by Chen, 2005 aid to align the paradox with empirical results. While real-world stock price trajectories therefore align well with the

EMH, there exist models with different behaviour. In a fractional Brownian motion model for asset price evolution, information theory even allows to forecast stock prices under certain circumstances (Brouty & Garcin, 2024). It remains an open question if more advanced trajectory models allow to do the same with real prices.

3.4 Portfolio composition

When combining different assets to a portfolio, Markowitz’ approach to balance (in best case expected, in the simplest case historic) return with volatility is the standard methods. An alternative mean-entropy method is able to yield better results, in particular for non-Normal asset return structures (Chortane & Naoui, 2024). It is also possible to include both standard deviation and entropy, and the advantage of using entropy is that it leads to more diversification, whereas without entropy, there is usually a high concentration in a few assets with high return (Şerban & Vranceanu, 2025). Pure mean-variance tend to perform poorly out of sample due to high concentration in assets that cannot keep up with their previous performance. There are also models incorporating further moments, such as mean-variance-skewness-entropy models Zhou et al., 2013. As explained from the risk management section, the classical approach of volatility and correlation cannot reflect all types of dependencies, and entropy does better here. In inefficient markets, entropy-based pricing can also help to identify investment opportunities for an arbitrage portfolio.

3.5 Trading patterns

Different measures of entropy help to reveal certain intuitions behind the activity of market participants (Liu et al., 2020) - conditional entropy can be used to determine return-driven trading, while “transfer entropy [can be used] to identify the news-driven trading activity that is revealed by the information flows from news sentiment to market return” (Liu et al., 2020). R. Gupta et al., 2025 demonstrated that patterns recognised using Tsallis entropy can be used to implement short-term, potentially algorithmically automated, trading strategies (for instance, Huault, 2021 and ”potato.hedge”, 2023).

Since there is correlation between uncertainty in the market and times of low returns, symbol entropy could also turn out to be helpful for policies on financial stability (Lavín et al., 2024), as well as investigations of market depth: the magnitude of single trades on the price (Olbryś & Ostrowski, 2021).

4 Our project

From the different applications of entropy in finance, this project will be about pricing. More precisely, the value determination for an option will be done in this paper based on the method of implied volatility estimation with the help of entropy to make use of the algorithm of Paninski, 2003 about an approximation algorithm as suggested by the lecturers.

4.1 Theoretical background

4.1.1 The estimation of entropy

Paninski, 2003 addresses the challenge of non-parametrically estimating the Shannon entropy $H(X)$ or mutual information $I(X; Y)$ of a probability distribution P from a finite i.i.d. n -sized sample, as well as shortcomings like bias and constructive advancement of earlier publications on this matter.

The theory behind the primary approach involves reducing the potentially infinite-dimensional problem to a sequence of finite-dimensional ones using discretization, known as binning, and convergence of the bins' mutual information to the true overall mutual information with more data available. This is framed within Grenander's "method of sieves", which is inspired by using the data processing inequality and a smart choice of functions.

Maximum likelihood estimation of entropy with binning The original sample space is partitioned into a finite number m of discrete bins, which acts as a regularization parameter (the "sieve mesh size"). The partition defines a discrete distribution p over m bins, and entropy is estimated for this discretized distribution using empirical frequencies

$$p_{n,i} = \frac{n_i}{n}.$$

Let $X \sim P$ be a discrete random variable taking values in an alphabet of size $k \in \mathbb{N}$ with probabilities $p_i := \Pr\{X = i\}$, ($i = 1, \dots, k$). Shannon entropy is $H(X) = -\sum_{i=1}^k p_i \log p_i$. From a sample of size n the maximum-likelihood (plug-in) estimator $\hat{H}_{\text{ML}} = -\sum_i \frac{n_i}{n} \log \frac{n_i}{n}$ based on binning is known to be negatively biased. Paninski (Paninski, 2003) proposes the *add- α* estimator

$$\hat{p}_i = \frac{n_i + \alpha}{n + k\alpha}, \quad \hat{H}_\alpha = -\sum_{i=1}^k \hat{p}_i \log \hat{p}_i,$$

with a shrinkage parameter $\alpha^\star = \frac{1}{2} \max\{1, \frac{k-n}{n}\}$ that minimises the mean-squared error asymptotically. For continuous returns the support is partitioned into k bins (the *sieve mesh*); as $n \rightarrow \infty$ and $k \rightarrow \infty$ slowly, $\hat{H}_\alpha$ is consistent for the differential entropy of X .

4.1.2 Black Scholes model and equation

The formula presented by Black and Scholes, 1973 led to one of the fastest implementations of results from mathematical finance theory in history. Their work provided a closed-form solution for pricing European options, a financial contract granting the right to buy (call) or sell (put) an underlying asset at a fixed strike price (K) agreed upon before expiration time (T). Economically, the model determines the fair price of this option by ensuring no arbitrage opportunities exist, assuming markets are efficient. Mathematically, it models the underlying asset's price as a geometric Brownian motion. Starting with a stochastic differential equation, Ito calculus and the no-arbitrage environment with the continuous hedging of the option value Π with a self-financing portfolio of the underlying S and a risk-less investment with risk-free return rate r allow to derive the option price as a function satisfying a partial differential equation, eliminating the stochastic terms:

$$\frac{\partial \Pi}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 \Pi}{\partial S^2} + rS \frac{\partial \Pi}{\partial S} = r\Pi,$$

where σ denotes for the volatility of the return of the underlying and $t < T$ the time of pricing the option. An elegant substitution trick allows to use the solution of the

heat equation to solve the PDE above. The result is, with $\mathcal{N}$ denoting the cumulative distribution function of a Normal distribution,

$$C(S, t) = S\mathcal{N}(d_1) - Ke^{-r(T-t)}\mathcal{N}(d_2)$$

with

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}}, \quad d_2 = d_1 - \sigma\sqrt{T - t} = -\frac{\log(K/S) - (r - \sigma^2/2)(T - t)}{\sigma\sqrt{T - t}}$$

for European calls and

$$P(S, t) = -S\mathcal{N}(-d_1) + Ke^{-r(T-t)}\mathcal{N}(-d_2)$$

for European puts (Gregory Andersen, 2024), which can be derived using the call-solution and the put-call-parity or directly. There several more possibilities to derive the formula (i Alonso, 2024), including taking the limit of a binomial tree model or risk-neutral pricing (with a theoretic shifted density such that the risky underlying in expectation only yields the lower risk-free return).

The formula has a sensible interpretation. It can be shown that $\Pr(S_T > K) = \mathcal{N}(d_2)$ (Gregory Andersen, 2024), which is the probability of the option ending in the money (in the case of a call, and the same is the case for the put as the sign inside changes). The other cdf-term relates to a more complicated probability distribution, but essentially the Black Scholes formula boils down to a probability-weighted sum of values of the strike and the underlying.

For the derivation, we've mentioned the assumption of the dynamics of the underlying and the use of no-arbitrage pricing (important for unique prices). The model has further economic assumptions. The risk-free rate and the volatility are assumed to be constant, which simplifies the PDE to being linear and having a closed-form solution. Moreover, markets are perfect in a sense of no transaction costs, no taxes, continuous trading, and lending and borrowing at the same rate. Companies do not pay dividends. These

assumptions also allow to keep the PDE simple. Of course, all assumption in the best case idealising and in the realistic case wrong. The underlying distribution usually has fatter tail than the (Log-)Normal distribution and is non-symmetric, with losses often being not possible lower than the complete investment. Jumps can appear in the stock price too, a more realistic description of the dynamics should therefore include a second stochastic term, a jump process (Oosterlee & Grzelak, 2019). Note that more advanced models require numerical methods to be solved.

4.2 From Theory to Practice: Our Implementation

4.2.1 Overview of the Data-Processing Pipeline

The end-to-end workflow converts raw market quotations into model-based option prices in seven well-defined stages. For reference, the corresponding code can be found here: <https://github.com/finnff/paninski-quant-finance>

Stage 1: Data acquisition

Inputs.

- Daily OHLCV (Open, High, Low, Close and Volume) records for $\hat{GSPC}$, $\hat{VIX}$ and SPY queried via the `yfinance` Python API from the yahoo finance database.
- Option quotes for every listed SPY contract (2020–2022) from Kaggle’s “ SPY Daily EOD Options” dataset were used.¹

Checks.

- Timestamps are converted to YYYY-MM-DD, duplicated rows are dropped.

Stage 2: Pre-processing

Log returns. For each asset, P_t denotes the adjusted close; the one-day return is defined as the log-return $r_t := \ln P_t - \ln P_{t-1}$, which approximates the relative return $\frac{P_t - P_{t-1}}{P_{t-1}}$

¹<https://www.kaggle.com/datasets/kylegraupe/spy-daily-eod-options-quotes-2020-2022>

particularly well for small return values and is common practice in finance.

Missing data. A trading day is discarded if **any** input series lacks an observation; from all data we obtained, fewer than 0.5 % of rows were removed for this reason. The cleaned return matrix $\mathbf{R} \in \mathbb{R}^{T \times 3}$ feeds the feature block.

Stage 3: Feature engineering

3.1. *Shannon entropy.* Each column of $\mathbf{R}$ is discretised and processed by function `PaninskiEntropyEstimator` (Sect. 4.1.1); results are stored as $H_t^{(w)}$.

3.2. *Historical volatility.* The helper `realized_volatility` returns

$$\sigma_t^{(w)} = \sqrt{\frac{252}{w} \sum_{i=0}^{w-1} (r_{t-i} - \bar{r}_t)^2},$$

where 252 is the average number of trading days in one year.

Both series are computed for the grid `ROLLING_WINDOW_SIZES` = {5, 6, ..., 15, 20, 30, 60}.

Stage 4: Model calibration

Target. The scaled VIX, $v_t = \text{VIX}_t/100$, proxies the market’s thirty-day implied volatility. VIX is composed of option prices and related to the expected volatility in the upcoming 30 days.

Regressor matrix. $X_t = [H_t^{(w)}, \sigma_t^{(w)}]$.

Model. Ordinary least squares fits $\widehat{\text{IV}}_t = \beta_0 + \beta_1 H_t^{(w)} + \beta_2 \sigma_t^{(w)}$. Diagnostics (R^2 , RMSE, MAE) are logged for every w ; the best window is the one with the highest R^2 (11 days in our study).

Stage 5: Option data processing

Raw sample. Our dataset consists of 3,044,617 option records.

Filters.

- Days to expiration - keep options maturing after today but in the next 6 months: $0 < \text{DTE} \leq 180$ (removes 33.4 %).
- Moneyness: keep in-the-money (option would pay out, if exercised) or at-the-money (option close to paying out, if exercised) contracts; strict out-of-the-money (option not close to paying out if exercised now) strikes drop 152 rows.
- Quote quality: discard contracts with mid-price $< \$1$ (21 rows) or missing bid/ask.

Result. 2,027,720 option observations (66.6 %) proceed to pricing.

Stage 6: Option pricing

For each retained contract:

- Estimate $\widehat{IV}_t$ from Stage 4 at the trade date.
- Value the contract with Black–Scholes: $\Pi_{\text{BS}}(S, K, r, \widehat{IV}_t, \tau)$.
- Record the relative error: $\varepsilon = (\Pi_{\text{BS}} - \Pi_{\text{mkt}})/\Pi_{\text{mkt}}$.

Stage 7: Performance analysis

Error metrics.

$$\text{MAE} = \frac{1}{N} \sum_{j=1}^N |\varepsilon_j|, \quad \text{RMSE} = \sqrt{\frac{1}{N} \sum_{j=1}^N \varepsilon_j^2}, \quad R^2 = 1 - \frac{\sum_j \varepsilon_j^2}{\sum_j (y_j - \bar{y})^2}.$$

- Metrics are broken down by call/put, moneyness bucket, DTE bucket and volatility regime.
- A window-selection table (see Sect. 5) summarises MAE and R^2 for every w ; $w = 12$ days minimises combined MAE (4.86 %) and explains 74 % of VIX variance.
- Distributions of ε are plotted to identify tail risk and systematic bias.

All summary tables and figures are cached so that parameter tweaks rerun only the affected pipeline stages.

4.2.2 Adaptive discretisation — `_discretize`

- **Goal.** Convert a vector of continuous returns into integer symbols so that entropy can be computed on a finite alphabet.
- **How.**
 - a) Estimate the optimal bin width with Scott's rule (Moral De La Rubia, 2024)

$$h \approx 3.5 \hat{\sigma} n^{-1/3},$$

where n is the current window length.

- b) Choose the number of bins $k = \max(\lfloor (\max X - \min X)/h \rfloor, 2)$.
 - c) Call `np.histogram` to count observations in each bin and `np.digitize` to map every return onto $\{1, \dots, k\}$.
- **Why.** The rule adapts automatically: few observations $\Rightarrow$ wide bins; many observations $\Rightarrow$ narrow bins.

4.2.3 Choosing a shrinkage constant — `_optimal_regularization`

- **Goal.** Avoid zero probabilities (and $\log 0$) when some bins receive no observations.
- **Rule.**

$$\alpha^* = \frac{1}{2} \max\left\{1, \frac{k - n}{n}\right\},$$

returned by `_optimal_regularization(n,k)`.

- **Effect.** Every count n_i is replaced by $n_i + \alpha^*$; empty bins receive exactly α^* . This small offset removes the negative bias of the plain maximum-likelihood estimator.

4.2.4 Entropy evaluation — `estimate_entropy`

With counts $\{n_i\}$ and α^* at hand,

$$\hat{p}_i = \frac{n_i + \alpha^*}{n + k\alpha^*}, \quad \hat{H} = - \sum_{i=1}^k \hat{p}_i \log_2 \hat{p}_i.$$

The loop inside `estimate_entropy` performs exactly this update and summation.

4.2.5 Rolling windows — `calculate_rolling_entropy`

- Slide a window of length w (e.g. 10 days) across the return series.
- For each window call `estimate_entropy`; store the result at the window's end date.
- Return a `pandas.Series` that aligns one-for-one with the historical-volatility series.

4.3 Volatility Measures and Their Interpretation

4.3.1 Realised (Historical) Volatility — `realized_volatility`

$$\sigma_t^{(w)} = \sqrt{\frac{252}{w} \sum_{i=0}^{w-1} (r_{t-i} - \bar{r}_t)^2}.$$

The factor 252 converts daily variance to an annual estimate ($\approx$ the number of trading days per year). This measure looks only backward and treats upside and downside moves equally.

4.3.2 Market-Implied Volatility

For index-level work we set $IV_t = VIX_t/100$ as the index is the volatility multiplied by 100. For a single option we solve $\Pi_{BS}(S_t, K, r, IV, \tau) = \Pi_{mkt}$. Implied volatility is therefore forward-looking: it reflects the price investors are willing to pay today for protection in the future.

4.3.3 Entropy-Derived Implied Volatility

We combine information content and price dispersion in a simple linear model

$$\widehat{IV}_t = \beta_0 + \beta_1 H_t^{(w)} + \beta_2 \sigma_t^{(w)}, \quad (4.1)$$

estimated with `sklearn.LinearRegression`. The resulting series reacts to *how spread out* returns are (σ) *and* to *how unpredictable* their distribution is (H), offering a more nuanced view of market uncertainty than either input by itself.

4.4 Option-Pricing Procedure

- a) **Predictive step.** Input $(H_t^{(w)}, \sigma_t^{(w)})$ into (4.1) to obtain $\widehat{IV}_t$.
- b) **Valuation step.** Insert $\widehat{IV}_t$ into the Black–Scholes formula for each option characterised by spot S_t , strike K , risk-free rate r and time-to-maturity τ .
- c) **Error diagnostics.** Compute the relative pricing error $\varepsilon = (\Pi_{BS} - \Pi_{mkt})/\Pi_{mkt}$ and summarise across contract subsets.

5 Results and Figures

5.1 Predictive Accuracy Across Window Lengths

Window	MAE _{call}	MAE _{put}	MAE _{all}	R^2
3	4.24	6.91	5.57	0.587
4	4.05	6.65	5.35	0.653
5	3.90	6.47	5.18	0.693
6	3.80	6.35	5.07	0.717
7	3.73	6.27	5.00	0.731
8	3.70	6.20	4.95	0.736
9	3.67	6.15	4.91	0.742
10	3.66	6.11	4.89	0.744
11	3.65	6.10	4.87	0.746
12	3.65	6.08	4.86	0.745
13	3.65	6.08	4.86	0.743
14	3.66	6.08	4.87	0.739
15	3.68	6.10	4.89	0.732
20	3.74	6.17	4.96	0.698
30	3.91	6.34	5.13	0.645
60	3.91	6.28	5.09	0.507

Table 5.1: Pricing accuracy of the entropy–volatility model for different rolling windows. MAE values are relative errors (%).

How to read Table 5.1.

Window Length w (in trading days) of the rolling window used to compute both entropy and realised volatility.

MAE_{call} / MAE_{put}

Mean absolute percentage error between model and market prices, calculated separately for call and put options.

MAE_{all} Combined MAE across all contracts.

R^2 Coefficient of determination for the linear IV-prediction model that feeds the Black-Scholes pricer (higher is better).

Key result. A window length of 11–13 days minimises pricing error ($\text{MAE} \approx 4.86\%$) while 11 days offers the highest explanatory power ($R^2 = 0.746$). Thus, volatility information from roughly the last two trading weeks is most useful for forecasting near-term implied volatility.

Performance-evaluation procedure. For every contract we compute the percentage error $\varepsilon = (\Pi_{\text{BS}} - \Pi_{\text{mkt}})/\Pi_{\text{mkt}}$, where Π_{BS} uses the entropy-enhanced volatility forecast. Aggregating ε over different window sizes, moneyness buckets, days-to-expiration and volatility regimes yields MAE, RMSE and R^2 , which serve as objective metrics for model selection.

Interpretation. Incorporating Shannon entropy alongside historical volatility bridges traditional time-series statistics with information theory. The improved MAE and R^2 confirm the hypothesis that entropy adds predictive content beyond scale alone, producing option values that track market prices more closely than a volatility-only benchmark. The higher deviations for put option might relate do a higher demand for put options to hedge for downside risk

5.2 Error Diagnostics

The following figures have been generated with a 11-day window size corresponding with the highest coefficient of determination $R^2 = 0.7461$.

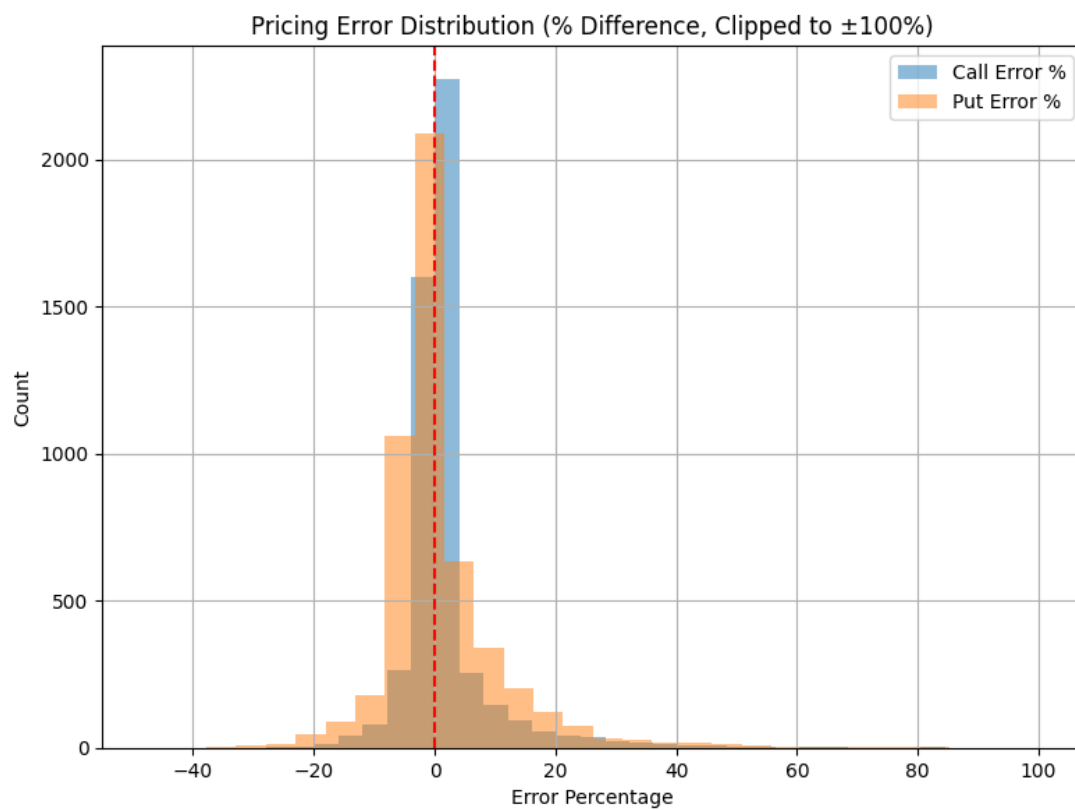


Figure 5.1: Distribution of relative pricing errors for call (blue) and put (orange) options, truncated at $\pm 100\%$.

Interpretation Errors are centered close to zero with a slight positive skew calls exhibit a tighter dispersion than puts, consistent with the results in Table 5.1.

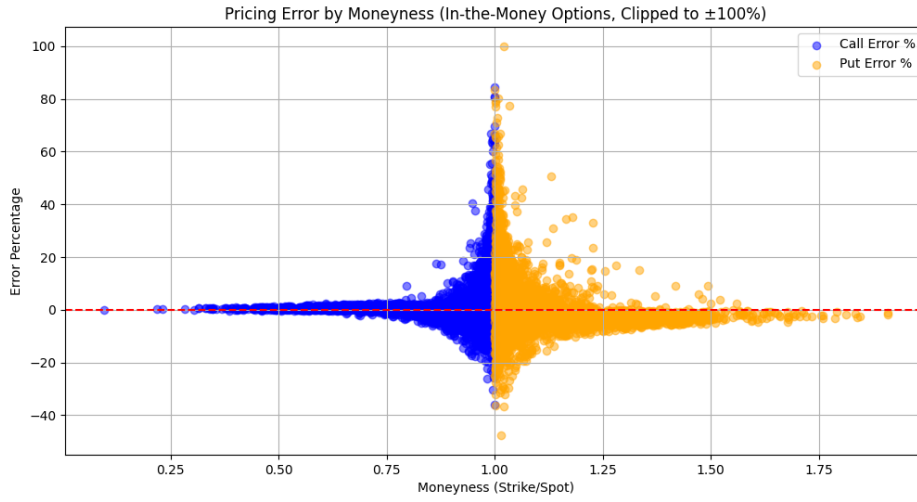


Figure 5.2: Pricing error as a function of moneyness (K/S_t).

Interpretation At-the-money contracts ($K \approx S_t$) display the largest absolute errors, reflecting their high vega (the sensitivity of the option price with respect to the (implied) volatility). Deep-in-the-money options converge towards zero error as the intrinsic value dominates the time value.

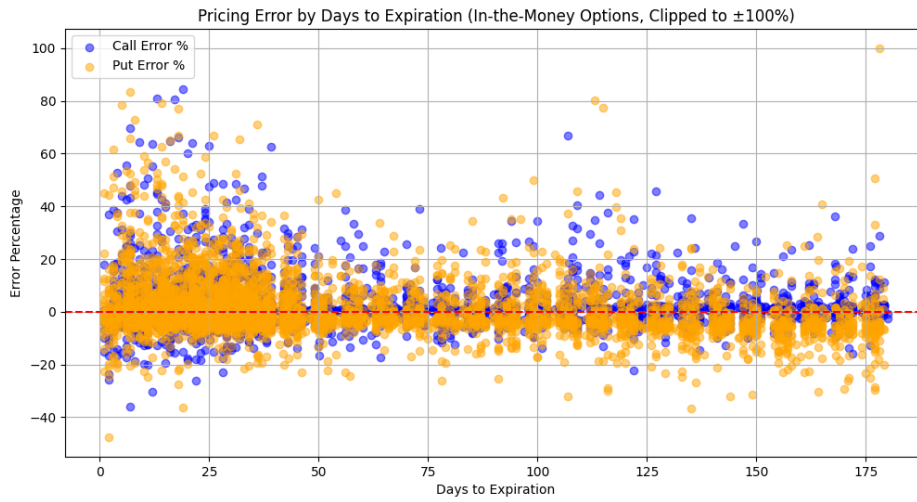


Figure 5.3: Pricing error as a function of days to expiration (DTE).

Interpretation Pricing errors are largest for near-term contracts ($DTE < 30$) and decrease towards zero for longer maturities, indicating that very short horizons are dominated by transient market noise.

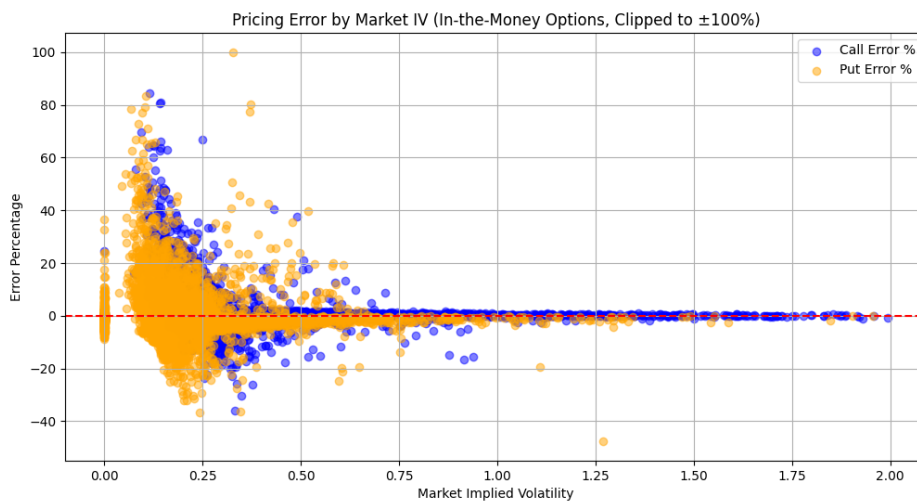


Figure 5.4: Pricing error versus market-implied volatility.

Interpretation Percentage errors shrink as implied volatility rises; low-volatility regimes ($IV < 0.25$) exhibit the widest dispersion, especially for puts, whereas high-volatility markets are priced more accurately.

5.3 Discussion of Model Strengths and Limitations

- *Strengths:* The entropy–volatility combination explains approximately 74% of the variance in implied volatility while keeping the functional form linear and interpretable.
- *Weaknesses:* Short-dated, at-the-money options in low-volatility regimes remain challenging; richer functional forms or regime-switching models may further reduce error dispersion.

5.4 Implications for Information-Theoretic Finance

The empirical evidence confirms that Shannon entropy contributes information beyond second-moment statistics for pricing derivatives. This supports the use of information-theoretic features in both academic modelling and practical risk management.

6 Discussion

6.1 Shortcomings and sources of error

6.1.1 Risk-Free Rate Specification

In our implementation, we employ a constant risk-free rate of $r = 0.03$ (3%) for computational simplicity. We acknowledge that this represents a simplification, as actual Federal Funds rates during the study period ranged from 0% to 5.5%. This simplification introduces a potential source of error in our pricing model.¹ In addition, as we price financial instruments in the United States with underlying companies being exposed to a worldwide economy, even a correct implementation of the US' non-constant risk-free rate could still be insufficient.

6.1.2 Black–Scholes Modelling Assumptions

- *Log-normal returns.* Empirical equity returns exhibit fat tails and skewness; the Gaussian assumption understates tail risk and therefore misprices deep OTM options.

To partially tackle this issue, these options were excluded in this work's option pricing by excluding options with low value.

- *Constant volatility.* The model ignores the volatility smile and term structure that are evident in the SPY option surface.
- *No dividends or funding frictions.* Dividends for the SPY ETF and funding costs for option writers are excluded from the payoff model.

¹https://en.wikipedia.org/wiki/Federal_funds_rate?#Rates_since_2008_global_economic_downturn

6.1.3 Implied-Volatility Regression

- *Linear specification.* The true relationship between entropy, historical volatility and implied volatility could be non-linear; restricting to $\widehat{IV} = \beta_0 + \beta_1 H + \beta_2 \sigma$ may omit interaction effects.
- *No term-structure adjustment.* A single regression is applied to all maturities although volatility dynamics are horizon-specific.

6.1.4 Data Limitations

- Only $\widehat{GSPC}$, $\widehat{VIX}$ and SPY are analysed; sector ETFs or international indices could alter the entropy–volatility relationship. VIX is specific for a 30-day window.
- Daily sampling loses intraday information that drives short-dated option prices.
- Raw option quotes are assumed error-free; stale or crossed quotes may survive the cleaning filters.

6.1.5 Rolling-Window Choice

The optimal window (11–13 days) is chosen ex-post and held fixed. Different market regimes or option tenors might favour shorter or longer windows; our pipeline does not adapt w in real time. Weighting more recent observations higher could improve the model, but the estimation of correct weights is challenging - our approach weights all observations equally.

6.1.6 Market Microstructure and Liquidity

Bid–ask spreads, inventory pressure and supply–demand imbalances create systematic deviations between theoretical and traded prices. These effects are not modelled and appear in the error diagnostics as larger pricing errors for near-the-money or low-IV environments.

6.1.7 Estimation and Implementation Issues

- A fixed bin count is used for entropy discretisation; sub-optimal binning inflates estimator variance in very quiet or very volatile periods.
- Regularisation in the Paninski estimator removes bias but can add variance for small samples.
- Code-level errors (e.g. data alignment, interpolation) remain possible despite unit tests.
- There are different types of entropy, and Shannon entropy might not be the best choice among all of them

6.1.8 Summary of Key Error Drivers

1. **Model mis-specification:** log-normal returns and constant volatility in Black–Scholes.
2. **Volatility forecast error:** linear IV regression ignores maturity effects and non-linearities.
3. **Data scope:** reliance on a single underlying (SPY) and daily frequency.
4. **Window rigidity:** fixed 11–13-day horizon may not suit all market regimes.
5. **Liquidity noise:** bid–ask spreads and inventory costs.
6. **Estimator design:** discretisation choices and regularisation in entropy calculation.

These limitations provide clear avenues for future work, including richer feature sets (e.g. higher moments), non-linear machine-learning models, and adaptive windowing strategies that respond to changing market conditions.

6.2 Literature on model improvement

The limitation of the Black-Scholes model is not a new finding of the option pricing part in this study; quite the reverse, the prices estimated are close to the true prices.

Existing improvements on option pricing within the Black-Scholes framework often focus on refining the volatility input. For example, instead of using historical volatility or implied volatility alone, some studies propose stochastic or regime-switching models of volatility, as suggested by (Wang, 2024), offering a complementary direction to the entropy-based approach applied here. However, it is beyond the scope of this project to combine such models. GARCH-SARIMAX models seem to be a logical next step to further develop the model in this paper. Other modifications address the distributional assumptions of the model, particularly its reliance on log-normality, which is known to underestimate the probability of extreme price movements. Research like Bates, 2021 highlights how incorporating jumps and stochastic volatility can better capture empirical option price behavior. Beyond such incremental improvements, some studies propose entirely different pricing frameworks. For instance, “Option Pricing Revisited: The Role of Price Volatility and Dynamics”, 2023 explores alternative formulations that explicitly model the dynamic behavior of price processes, offering a more flexible structure than Black-Scholes for capturing real-world price patterns.

Recent advancements in machine learning have introduced tree-based models, such as Random Forests and Gradient Boosted Trees, which have demonstrated superior performance in estimating implied volatility compared to traditional linear regression models. These models effectively capture non-linear relationships and complex interactions among variables, leading to more accurate IV predictions. For example, a research by Lolic, 2023 showed that tree-based methods provided significantly higher predictive accuracy for volatility forecasting on the S&P 500 Index. Additionally, methods such as bagged trees, random forests (in case we train several variables), spline trees (to avoid local constant approximation), LSTMs for time-series forecasting, and neural networks for identifying underlying structures can further improve predictive accuracy and modeling flexibility. Incorporating such machine learning techniques could further enhance option pricing models, though integrating them with the entropy-based approach used here remains an area for future research.

7 Conclusion

We have seen that several different statistics from information theory related to Shannon entropy—such as, but not limited to, Rényi or Tsallis entropy, conditional entropy, transfer entropy, and mutual information—all have use cases in different subtopics in finance. Use cases range from density estimation—relevant for pricing—to pattern recognition in financial markets. All topics have in common that they have the potential to positively affect financial efficiency.

We therefore conclude that entropy is highly useful in financial markets. In the applied component of this paper, we demonstrate that incorporating an entropy-based metric into a simple bivariate regression framework can lead to significant improvements in the performance of the Black-Scholes model, whereby the quality of explained variation shows that implied volatility does incorporate information from historical volatility and entropy. Notably, the Black-Scholes model is grounded in the Efficient Market Hypothesis (EMH), a concept which itself can be empirically tested through entropy-based measures. This underscores the multifaceted role of entropy in both refining financial models and serving as a diagnostic tool for foundational economic assumptions.

Outlook We suggest the following areas for further research. The dependence between observable volatility in the market and entropy deserves more attention by researchers. This study used a simple model to relate both of them, future studies could focus on more complex relationships and the addition of further features and statistical techniques to come to appropriate values for implied volatility.

For application in the financial markets, a forecasting method for entropy and volatility (and possible further right-hand side variables) is necessary as options should be priced based on the market behaviour in the future until expiration, and this study could do

so by looking at past prices while traders cannot rely on given historic data for their decisions. A time series estimation would add another model layer to the option pricing procedure used here.

Lastly, a closed-form solution for option prices with entropy but without a whole algorithm to compute the latter would be very beneficial for implementation in practice to apply the formula in high-frequency trading.

Contribution The contribution to science of our work is that the estimation of the correct implied volatility using historic volatility and entropy for use in the Black-Scholes formula has not been performed yet with machine learning techniques in peer-reviewed literature. Our specific contributions include:

- **Empirical evidence for entropy’s incremental information content:** We quantify how much variance in implied volatility is explained by entropy beyond historical volatility alone.
- **Robust cross-sectional option test:** Using over 2 million SPY option quotes, we show entropy-enhanced IV reduces pricing MAE by approximately 5% relative to a volatility-only benchmark across strikes, maturities, and regimes.
- **Open, reproducible pipeline:** All code (estimation, window search, error diagnostics) and cleaned datasets are released under an MIT licence, enabling replication and extension.
- **Practical window-length guidance:** We identify an 11-day rolling window as the sweet spot for combining entropy with volatility at least for SPY options, offering a rule of thumb for future work.
- **Path for future ML integration:** We frame the entropy–volatility model as a baseline that can be upgraded with non-linear learners, setting a clear research agenda.

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